

EDUCATION

Carnegie Mellon University <i>M.S. in Software Engineering GPA: 4.0 / 4.0</i> Selected Coursework: Distributed Systems, Software Refactoring, Machine Learning, LLM Systems	Expected May 2027
University of Wisconsin–Madison <i>B.S. in Computer Science & B.S. in Computer Engineering GPA: 3.72 / 4.0</i> Selected Coursework: Algorithms, Operating Systems, Software Engineering, Computer Architecture	May 2024

WORK EXPERIENCE

NewsBreak <i>Software Engineering Intern</i>	May – August 2026 <i>Mountain View, CA</i>
- Built the FastAPI / async MongoDB backend from its first commit to 89 REST endpoints across 27 routers and 7 Kubernetes CronJobs on AWS EKS, with ~1,450 fully offline tests (1:1 with production lines, 0.17 s) behind a ruff / pyright / pytest CI gate. - Sole SwiftUI engineer and primary backend author on SportsBreak (~6,000 installs, 1,119 peak daily active users, 1.4K App Store ratings); authored 344 PRs across five repos (340 merged). - Replaced a 25–30 request client fan-out with one backend call; redesigned a cache that could not hit (13.2 MB keyed per user) into a shared 10.9 MB catalog cache, 78× smaller and sized for a 512 MiB pod. - Designed a personalized ingest → embed → LLM-score → dedupe → push pipeline; calibrated on 4,100+ production-shadow decisions, it cut filler alerts 34% → 4% and held its 2/day cap over 11,745 user-days. - Designed an agentic search API: one SSE endpoint, deterministic RAG on turn 1 and a bounded tool-calling agent after; entity resolution took real logged queries from 0% to 100% P@1.	
Haddee Education <i>Engineering Lead</i>	June – September 2025 <i>San Diego, CA</i>
Led an 8-intern team through Agile sprints to ship a React, Next.js, FastAPI, and Supabase web app from 0→1 on GCP App Engine; designed the schemas and REST APIs and led PR reviews.	
ZenAI — MiraclePlus (Formerly YC China) 2023 Fall Batch <i>Full Stack Developer</i>	September – December 2023 <i>Beijing, China</i>
- Built a VC-backed startup’s LLM mock-interview web app from 0→1 as the sole engineer in React, Next.js, Supabase, and PostgreSQL; owned auth, session state, and database interactions. - Demoed the product and iterated with clients until three 200–500-person companies signed contracts.	

RESEARCH EXPERIENCE

Control & Learning Group — Carnegie Mellon University <i>Graduate Researcher Advisor: Prof. Yorie Nakahira</i>	February – June 2026
- Developed an Online Adaptive Probabilistic Safety Certificate (PSC) with PhD researchers; outperformed Adaptive MPC in 3-DOF vehicle simulations under environmental uncertainty. - Mapped natural-language instructions to formal safety specs with multi-turn LLMs and implemented Bayesian estimators for real-time road-friction adaptation (LuGre tire models).	

SKILLS

Languages	Python, TypeScript, JavaScript, Java, Go, Swift, SQL, C, Bash
Frameworks	FastAPI, Pydantic, React, Next.js, SwiftUI, Express, LangChain, Firebase, Supabase
Data & AI	MongoDB, PostgreSQL, MySQL, Redis Streams, PyTorch, RAG, LLM agents, Langfuse
Cloud & DevOps	AWS, GCP, Kubernetes (EKS), Docker, Helm, Jenkins, GitHub Actions, Prometheus
Practices	REST APIs, pytest, Unit Testing, CI/CD, Agile/Scrum, Git, Jira
Certifications	AWS Certified Solutions Architect – Associate, AWS Certified Cloud Practitioner

PROJECTS & LEADERSHIP

- my-av:** Built a PyTorch CNN-GRU that predicts steering angle and speed from 20-frame (1s) video; processed 33 hours of Comma2k19 and trained on local CUDA and GCP Vertex AI.
- ECE Leadership & Engagement Fellow** (CMU, Aug 2026 – present): Own budgeting, marketing, and outreach for ECE student programs — company-sponsored hackathons, alumni events, and tech event outings.